

Saniya Jaswani (9 years exp)

7 years in Data Scientist | 2 years in Software Engineering

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SUMMARY-

Data Scientist 7 yrs of exp in machine learning, NLP, and deep learning, leading Gen AI initiatives. Built scalable LLM multi agents system, recommendation systems to improve sales rep targets and inventory planning for retail sector. 2 yr as software engineer developing application using Rest API, Angular , React & .Net

PROFESSIONAL EXPERIENCE

Generative AI Data Scientist | ACQUIA

Mar 2024 - Nov 2024

- **Search AI Assistant** - Developed a search agent that enables intelligent retrieval and summarization of documents using file content, metadata, and embeddings. Integrated PGVector for vector storage, *LLMs* (Claude, Gemini) for semantic search and summarization, and Multi agent using *LangChain* , *LangGraph* . Significantly improved user experience by automating deep file search, content extraction, and Q&A over PDFs, DOCs, and text files.

Senior Data Scientist | ZS Associates

Apr 2022 - Dec 2023

- **Recommendation System** - Developed a recommendation system to optimize multi-channel engagement for sales rep promoting drug products to HCPs. Leveraged PySpark to efficiently process large-scale sales and interaction data, and implemented a Neural Collaborative Filtering model to predict HCP engagement across channels. The system contributed to an 8% increase in monthly sales (from a \$100K baseline), demonstrating measurable impact in a high-value domain.
- **Market Mix Modelling** - Implemented a MMM solution to evaluate the ROI and sales contribution of various media channels used in pharmaceutical campaigns. Insights from the model enabled data-driven budget optimization, enhancing future media planning and improving overall marketing efficiency.
- **Optimizing Budget** - Built a budget optimization model to allocate marketing spend across channels for maximum ROI. Applied optimization algorithms (non linear programming) to simulate various budget scenarios several constraints. Resulting in improved marketing efficiency and optimal allocation of multi-million dollar budgets.
- **Inventory planning** - Designed and implemented model with demand forecasting using **LSTM** achieving low RMSE and high predictive accuracy. Automated the entire pipeline via Databricks Workflows using Spark.

Data Scientist | IBM

Aug 2018 - Mar 2022

- Customer Profiling - Built a customer profiling system using unsupervised learning (**Clustering techniques**) to segment users. Aiming to enhanced personalized recommendations and marketing strategies, leading to improved engagement and higher campaign conversion rates.
- Developed an automated performance monitoring system using **Tableau** that tracked application KPIs and generated weekly health reports for stakeholders
- **Product Driverless forecasting** - Designed a driverless forecasting engine combining XGBoost, GRU, RNN, and statistical models (ARIMA, ETS) into an ensemble pipeline for robust, accurate predictions across time series use cases. Aiming to automate process of forecasting and faster access to results.

Software Engineer | Infosys

July 2015 - Aug 2018

- Developed scalable REST APIs using Python and FastAPI to showcase Saving application for financial clients.
- End to end application development of software for Saving and current account journey of customer using . Net, Angular, SQL.
- Utilized **AWS** services including Sagemaker , S3, Lambda, and CloudWatch to deploy scalable applications, ensuring high
- availability, automated monitoring.

EDUCATION

Paper accepted on **CVPR** workshop 2025 on GenPhoto- a Visual interactive system for TTI generation

Master of Technology in AI from IIT Jodhpur

Bachelor in Computer Science from IET DAVV

Jan 2021 - Dec 2024

May 2011 - May 2015

SKILLS

Gen AI : GPT , Claude , LLM , RAG , Langchain , Prompt Tuning, Hugging face, Business applications of AI ,Automated content generation, content understanding, OpenAI, Anthropic, Cohere.

Deep Learning : BERT, LSTM , Neural Network, NLP, GRU , RNN, Natural Language Processing

Machine learning : Timeseries, Boosting, Random forest, Regression, Recommendation , MLOPS, Optimization, Classification

Technology : Python , GIT, Cuda, Sklearn , Pytorch , Tensorflow, SQL, Statistics , Pyspark, Hadoop, Tableau, AWS , Spark, Docker, AirFlow, A/ B testing , Spark ML , Data Engineering .

PROJECTS

Conversational AI Assistant - LLM Models , RAG , LangChain , AWS bedrock

- Application to enable quicker and natural search using LLM by user , and search across different domains. Worked with different Claude models, created a workflow of request to breakdown the query and navigate it through complex existing APIs
- Worked in AWS sagemaker , s3 , cloudwatch and EC2 to develop and deploy the system.

Recommender system

- Developed system to help pharma client target right customer with multiple channel and right time, used Neural Collaboration Filtering to understand complexity with multiple channels.
 - Creating customer profiling to understand deeper features of customer before adding any recommendations.
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